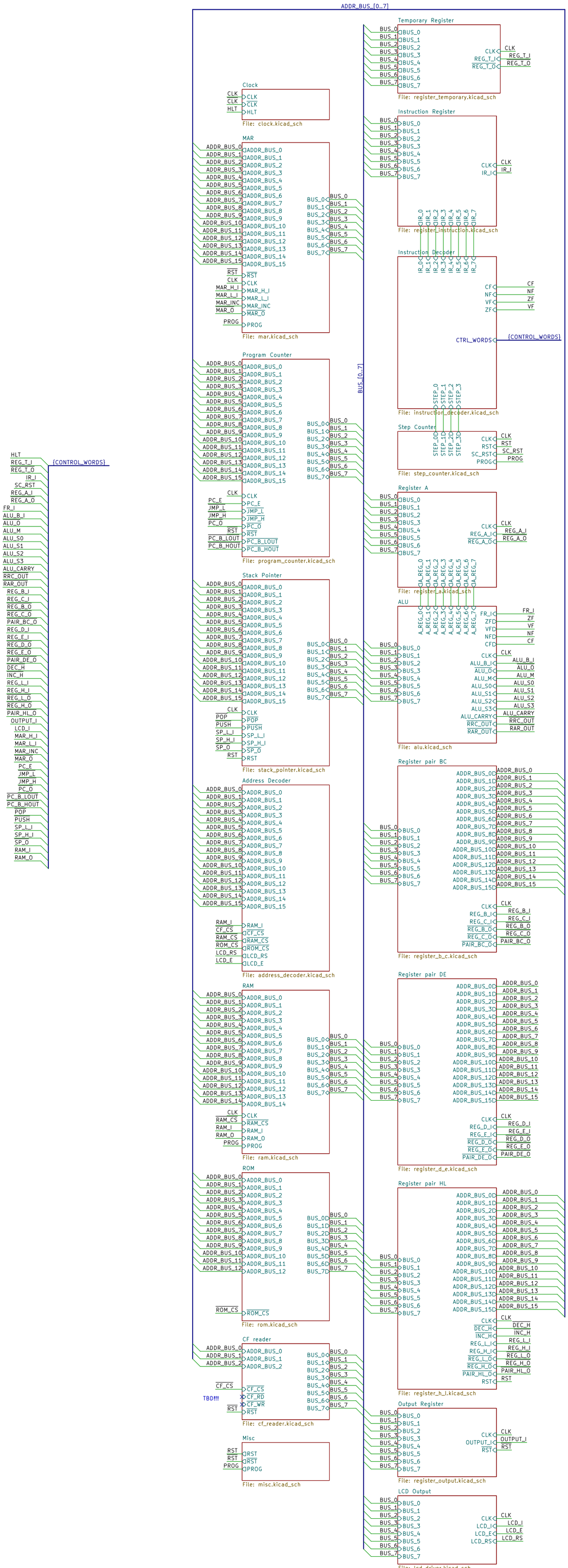
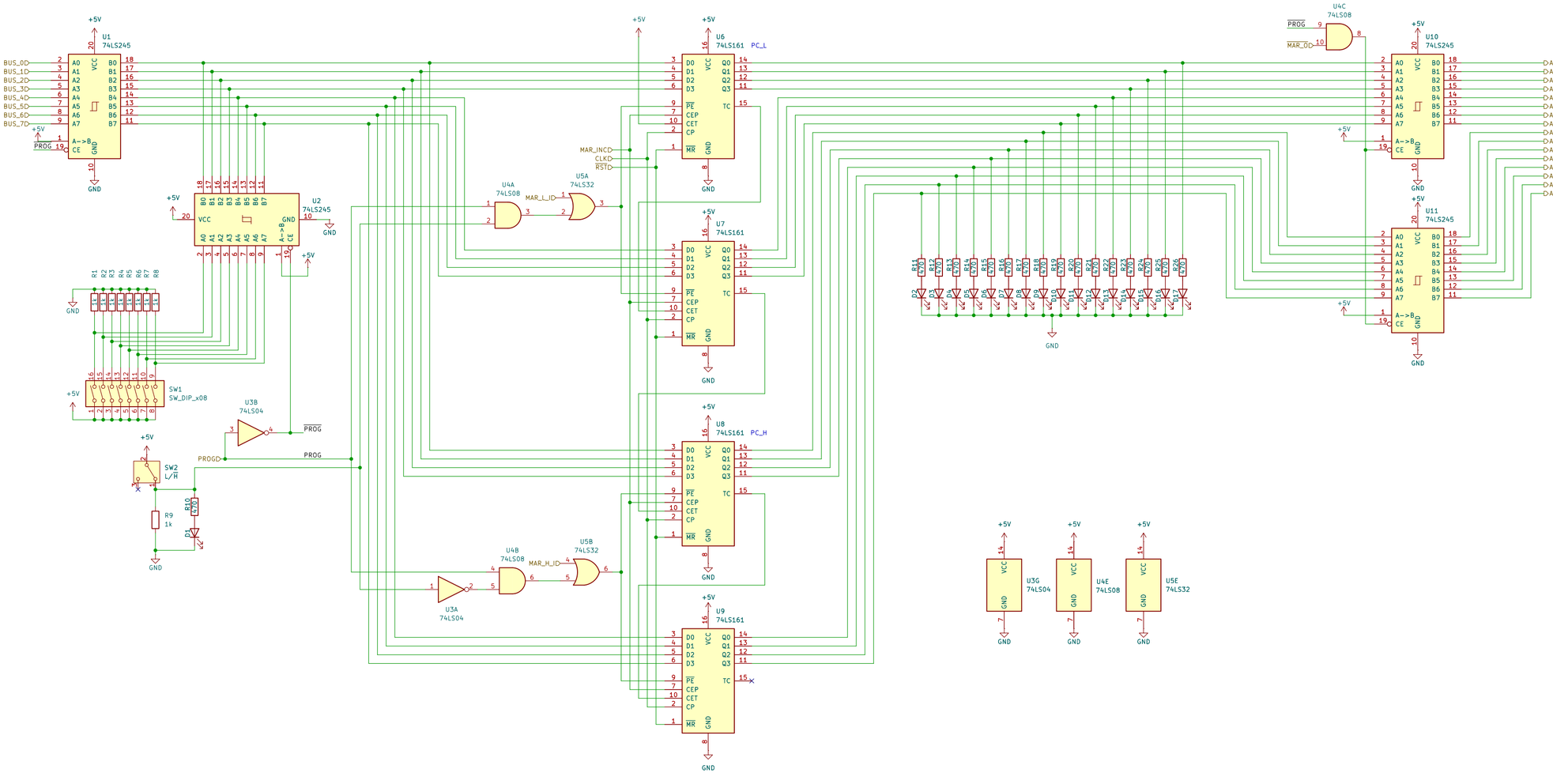


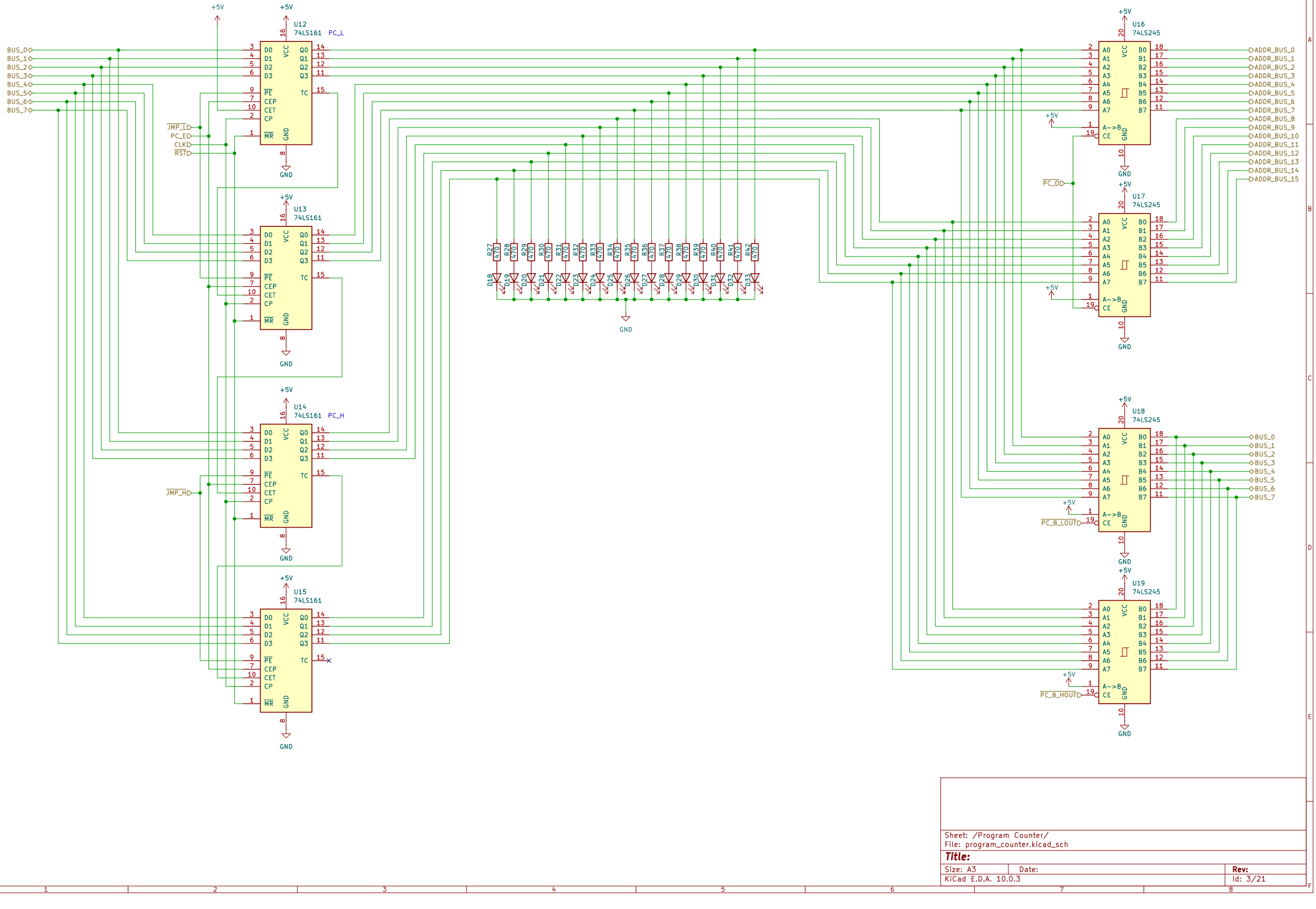
Sheet: /		
File: 8bit.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 1/21



Sheet: /MAR/
File: mar.kicad_sch

Title:

Size: A2 | Date: | **Rev:**
KiCad E.D.A. 10.0.3 | Id: 2/21



Sheet: /Program Counter/
File: program_counter.kicad_sch

Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.3 | Id: 3/21

BUS_0
BUS_1
BUS_2
BUS_3
BUS_4
BUS_5
BUS_6
BUS_7

U21
74LS193
U22
74LS193
U23
74LS193
U24
74LS193
+5V
GND
A
B
C
D
QA
QB
QC
QD
LOAD
CO
BO
UP
DOWN
CLR
VCC

POP
PUSH
RST

U20A
74LS00
SP_L_ID
CLK

U20B
74LS00
SP_H_ID

U20E
74LS00

R43 470 D34
R44 470 D35
R45 470 D36
R46 470 D37
R47 470 D38
R48 470 D39
R49 470 D40
R50 470 D41
R51 470 D42
R52 470 D43
R53 470 D44
R54 470 D45
R55 470 D46
R56 470 D47
R57 470 D48
R58 470 D49

U25
74LS245
U26
74LS245
A0
A1
A2
A3
A4
A5
A6
A7
B0
B1
B2
B3
B4
B5
B6
B7
A->B
CE

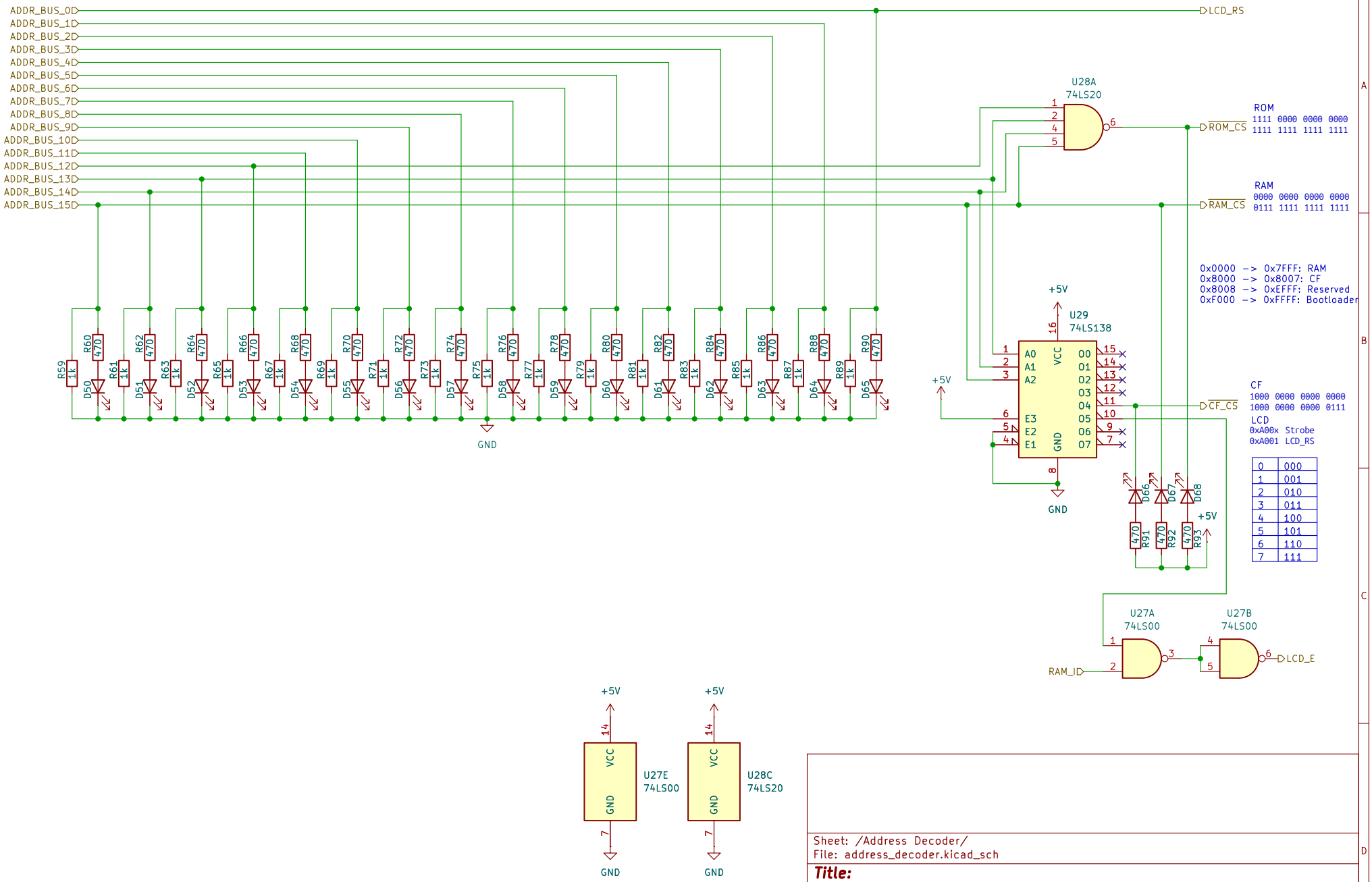
SP_0

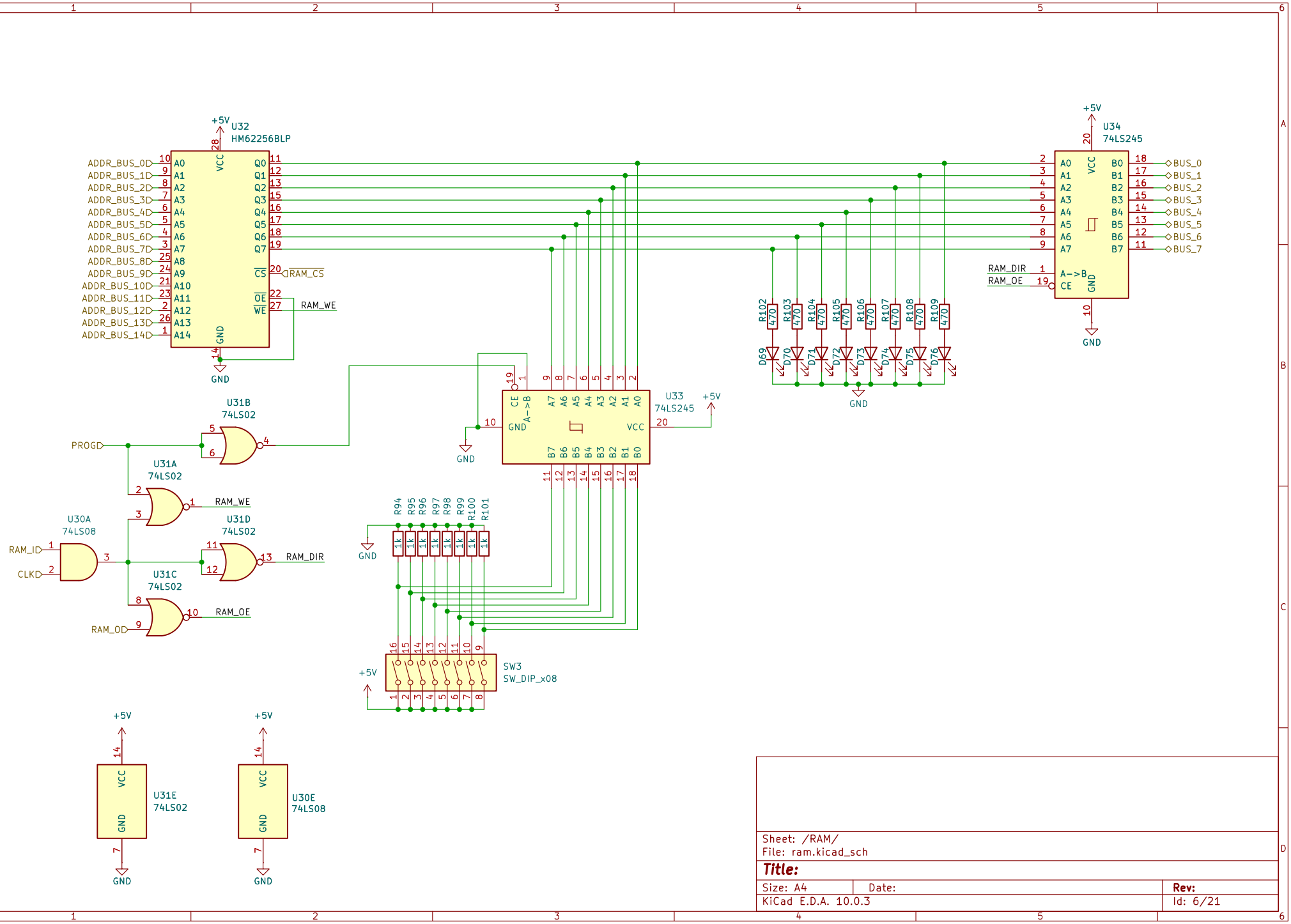
ADDR_BUS_0
ADDR_BUS_1
ADDR_BUS_2
ADDR_BUS_3
ADDR_BUS_4
ADDR_BUS_5
ADDR_BUS_6
ADDR_BUS_7
ADDR_BUS_8
ADDR_BUS_9
ADDR_BUS_10
ADDR_BUS_11
ADDR_BUS_12
ADDR_BUS_13
ADDR_BUS_14
ADDR_BUS_15

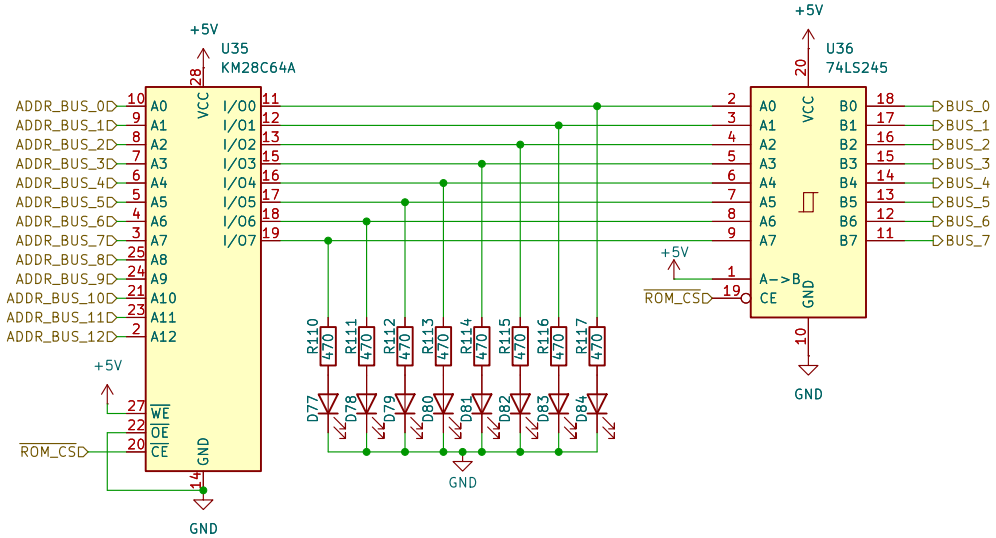
Sheet: /Stack Pointer/
File: stack_pointer.kicad_sch

Title:

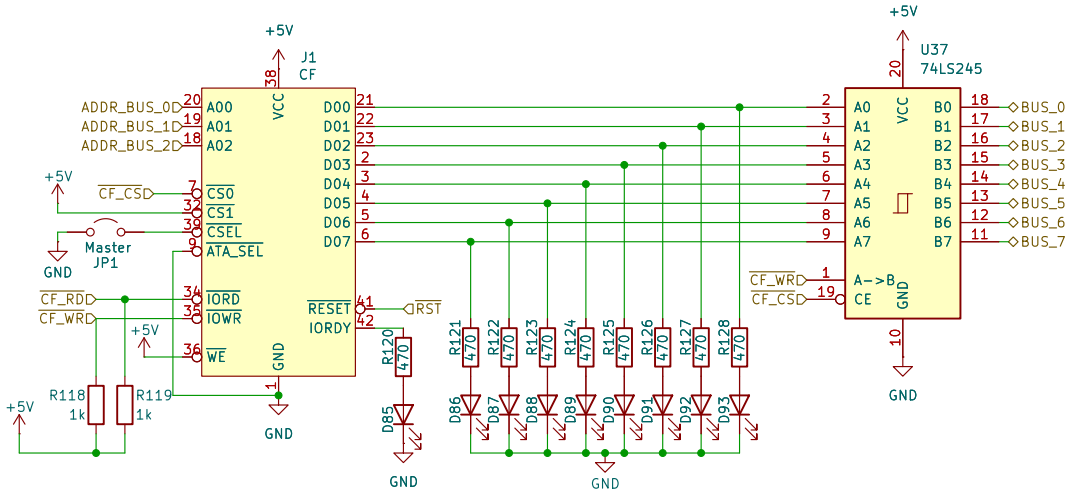
Size: A3	Date:	**Rev:**
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ADDR_BUS_0
ADDR_BUS_1
ADDR_BUS_2
ADDR_BUS_3
ADDR_BUS_4
ADDR_BUS_5
ADDR_BUS_6
ADDR_BUS_7
ADDR_BUS_8
ADDR_BUS_9
ADDR_BUS_10
ADDR_BUS_11
ADDR_BUS_12
ADDR_BUS_13
ADDR_BUS_14
ADDR_BUS_15
LCD_RS
U28A
74LS20
ROM
1111 0000 0000 0000
1111 1111 1111 1111
ROM_CS
RAM
0000 0000 0000 0000
0111 1111 1111 1111
RAM_CS
0x0000 -> 0x7FFF: RAM
0x8000 -> 0x8007: CF
0x8008 -> 0xEFFF: Reserved
0xF000 -> 0xFFFF: Bootloader
+5V
U29
74LS138
A0
A1
A2
E3
E2
E1
VCC
GND
O0
O1
O2
O3
O4
O5
O6
O7
CF
1000 0000 0000 0000
1000 0000 0000 0111
CF_CS
LCD
0xA00x Strobe
0xA001 LCD_RS
0 000
1 001
2 010
3 011
4 100
5 101
6 110
7 111
U27A
74LS00
U27B
74LS00
RAM_ID
LCD_E
U27E
74LS00
U28C
74LS20
Sheet: /Address Decoder/
File: address_decoder.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
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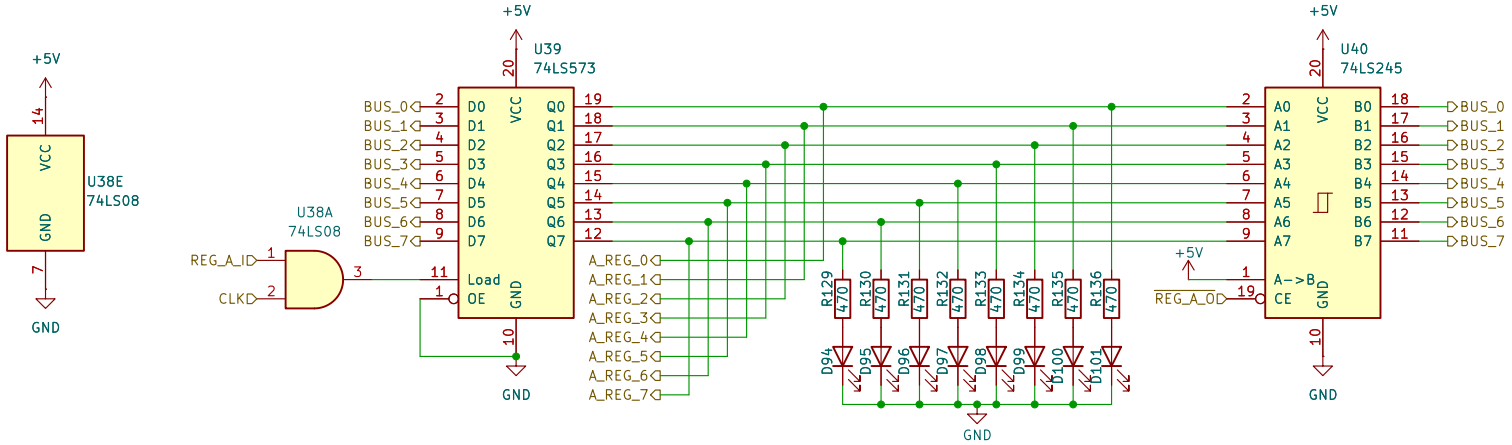

U32
HM62256BLP
+5V
GND
ADDR_BUS_0
ADDR_BUS_1
ADDR_BUS_2
ADDR_BUS_3
ADDR_BUS_4
ADDR_BUS_5
ADDR_BUS_6
ADDR_BUS_7
ADDR_BUS_8
ADDR_BUS_9
ADDR_BUS_10
ADDR_BUS_11
ADDR_BUS_12
ADDR_BUS_13
ADDR_BUS_14
RAM_CS
RAM_WE
U34
74LS245
RAM_DIR
RAM_OE
BUS_0
BUS_1
BUS_2
BUS_3
BUS_4
BUS_5
BUS_6
BUS_7
R102
R103
R104
R105
R106
R107
R108
R109
470
D69
D70
D71
D72
D73
D74
D75
D76
U33
74LS245
U31B
74LS02
PROG
U31A
74LS02
RAM_WE
U30A
74LS08
RAM_IO
CLK
U31D
74LS02
RAM_DIR
U31C
74LS02
RAM_OE
RAM_OD
R94
R95
R96
R97
R98
R99
R100
R101
1k
SW3
SW_DIP_x08
U31E
74LS02
U30E
74LS08
Sheet: /RAM/
File: ram.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
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Sheet: /ROM/		
File: rom.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 7/21



Sheet: /CF reader/		
File: cf_reader.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 8/21



Sheet: /Register A/		
File: register_a.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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BUS_0
BUS_1
BUS_2
BUS_3
BUS_4
BUS_5
BUS_6
BUS_7

U41A
74LS00
CLK
REG_L_ID

U42
74LS193
INC_H
DEC_H
RST
investigate negated behaviour

U43
74LS193

U44
74LS193

U41B
74LS00
CLK
REG_H_ID

U45
74LS193

U41E
74LS00

R137 470 R139 470 R141 470 R143 470 R145 470 R147 470 R149 470 R151 470
D102 D104 D106 D108 D110 D112 D114 D116

R138 470 R140 470 R142 470 R144 470 R146 470 R148 470 R150 470 R152 470
D103 D105 D107 D109 D111 D113 D115 D117

U48
74LS245
REG_L_OD

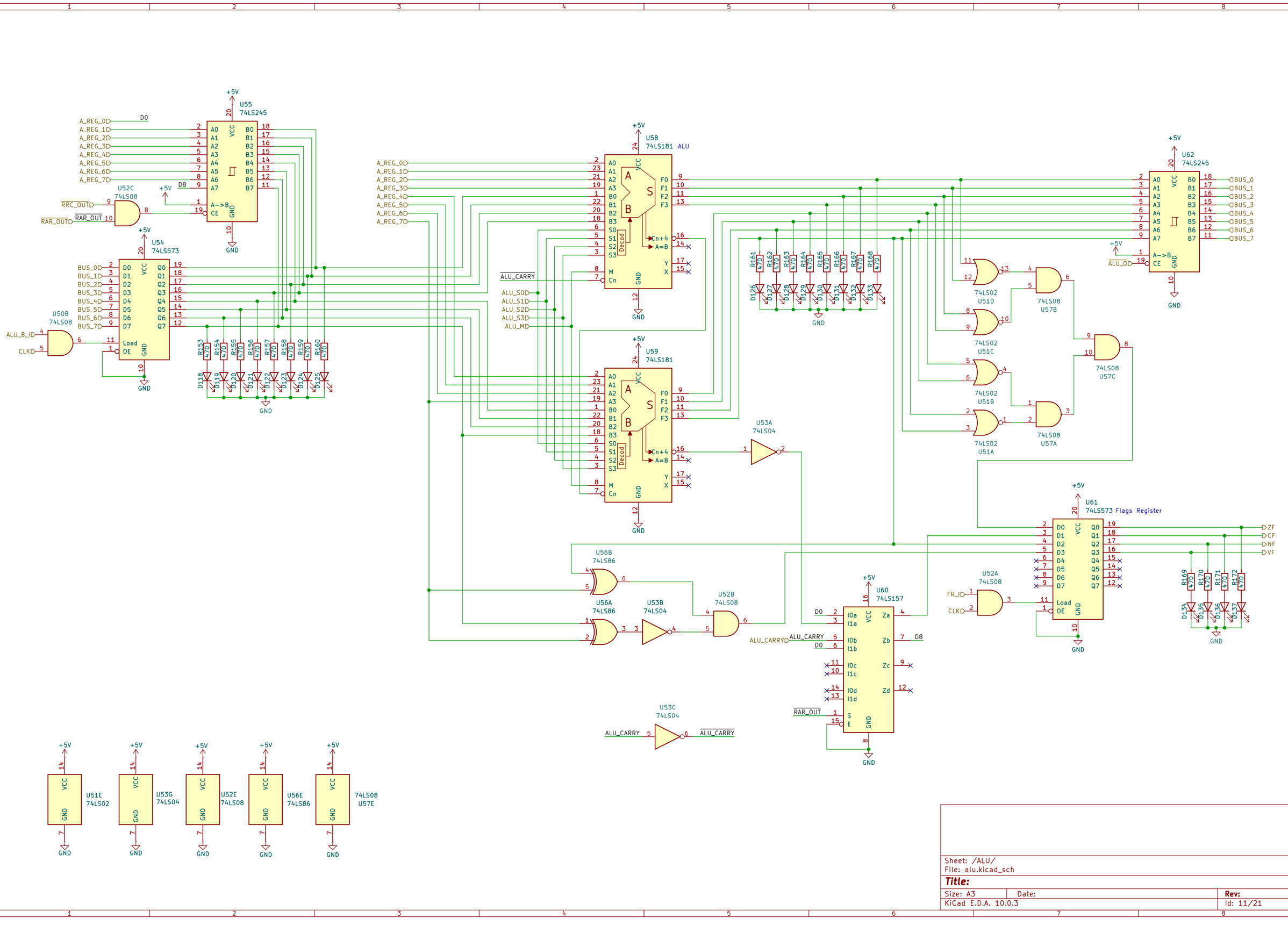
U49
74LS245
REG_H_OD

U46
74LS245
PAIR_HL_OD

U47
74LS245

ADDR_BUS_0
ADDR_BUS_1
ADDR_BUS_2
ADDR_BUS_3
ADDR_BUS_4
ADDR_BUS_5
ADDR_BUS_6
ADDR_BUS_7
ADDR_BUS_8
ADDR_BUS_9
ADDR_BUS_10
ADDR_BUS_11
ADDR_BUS_12
ADDR_BUS_13
ADDR_BUS_14
ADDR_BUS_15

Sheet: /Register pair HL/
File: register_h_l.kicad_sch
Title:
Size: A3 | Date: | **Rev:**
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Sheet: /ALU/
File: alu.kicad_sch

Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 11/21

BUS_0
BUS_1
BUS_2
BUS_3
BUS_4
BUS_5
BUS_6
BUS_7

U64
74LS573
U65
74LS573
U66
74LS245
U67
74LS245
U68
74LS245
U69
74LS245
U63A
74LS08
U63B
74LS08
U63E
74LS08

+5V
GND
REG_B_ID
CLK
REG_C_ID
REG_B_OE
REG_C_OE
PAIR_BC_OE

R173 R175 R177 R179 R181 R183 R185 R187
R174 R176 R178 R180 R182 R184 R186 R188
470
D138 D140 D142 D144 D146 D148 D150 D152
D139 D141 D143 D145 D147 D149 D151 D153

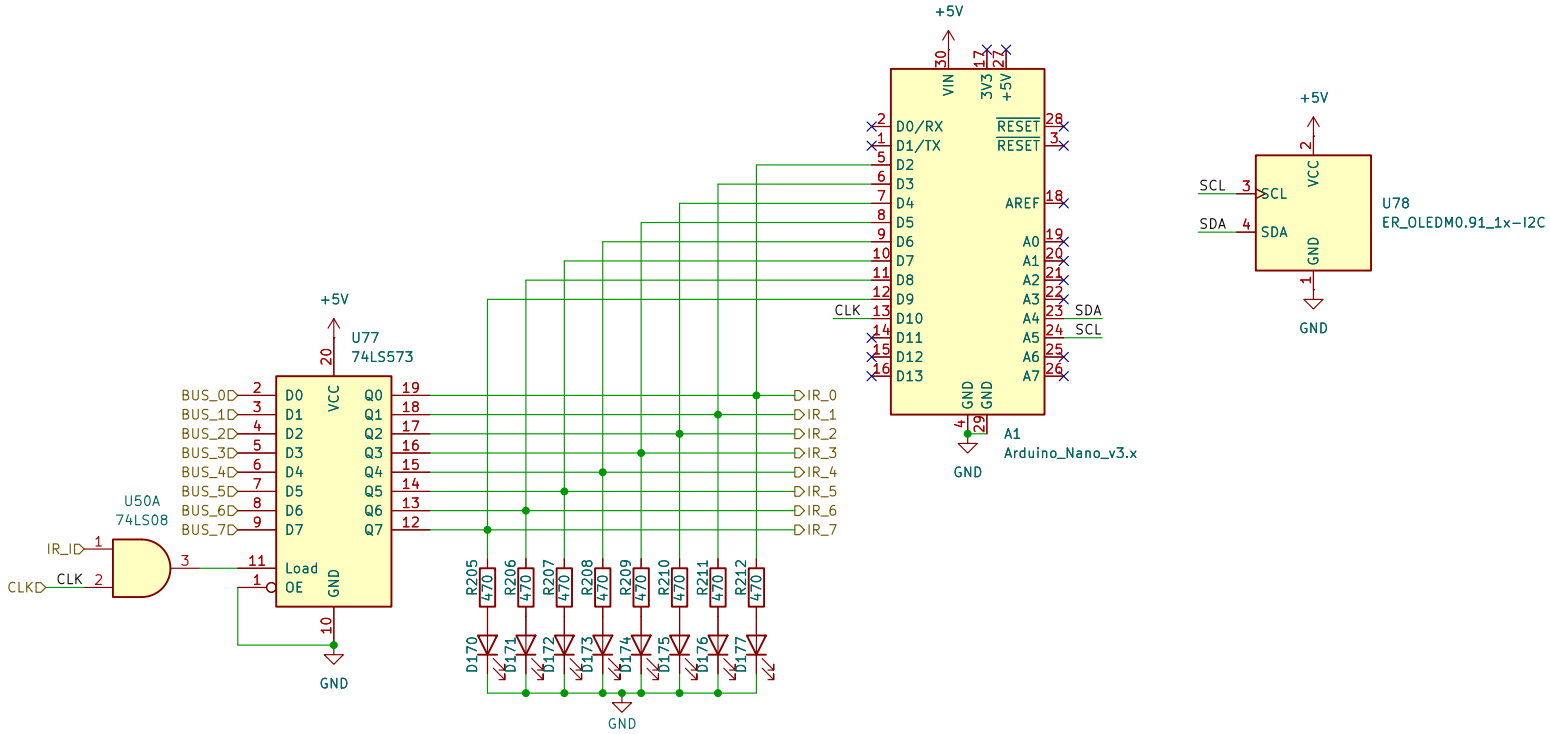
ADDR_BUS_0
ADDR_BUS_1
ADDR_BUS_2
ADDR_BUS_3
ADDR_BUS_4
ADDR_BUS_5
ADDR_BUS_6
ADDR_BUS_7
ADDR_BUS_8
ADDR_BUS_9
ADDR_BUS_10
ADDR_BUS_11
ADDR_BUS_12
ADDR_BUS_13
ADDR_BUS_14
ADDR_BUS_15

Sheet: /Register pair BC/
File: register_b_c.kicad_sch
Title:
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.3 | Id: 12/21

BUS_0
BUS_1
BUS_2
BUS_3
BUS_4
BUS_5
BUS_6
BUS_7

U71
74LS573
U72
74LS573
U73
74LS245
U74
74LS245
U75
74LS245
U76
74LS245
U70A
74LS08
U70B
74LS08
U70E
74LS08
+5V
GND
REG_D_ID
CLK_D
REG_E_ID
CLK_E
REG_D_OD
REG_E_OD
PAIR_DE_OD
R189 R190 R191 R192 R193 R194 R195 R196 R197 R198 R199 R200 R201 R202 R203 R204 470
D154 D155 D156 D157 D158 D159 D160 D161 D162 D163 D164 D165 D166 D167 D168 D169 LED
ADDR_BUS_0
ADDR_BUS_1
ADDR_BUS_2
ADDR_BUS_3
ADDR_BUS_4
ADDR_BUS_5
ADDR_BUS_6
ADDR_BUS_7
ADDR_BUS_8
ADDR_BUS_9
ADDR_BUS_10
ADDR_BUS_11
ADDR_BUS_12
ADDR_BUS_13
ADDR_BUS_14
ADDR_BUS_15

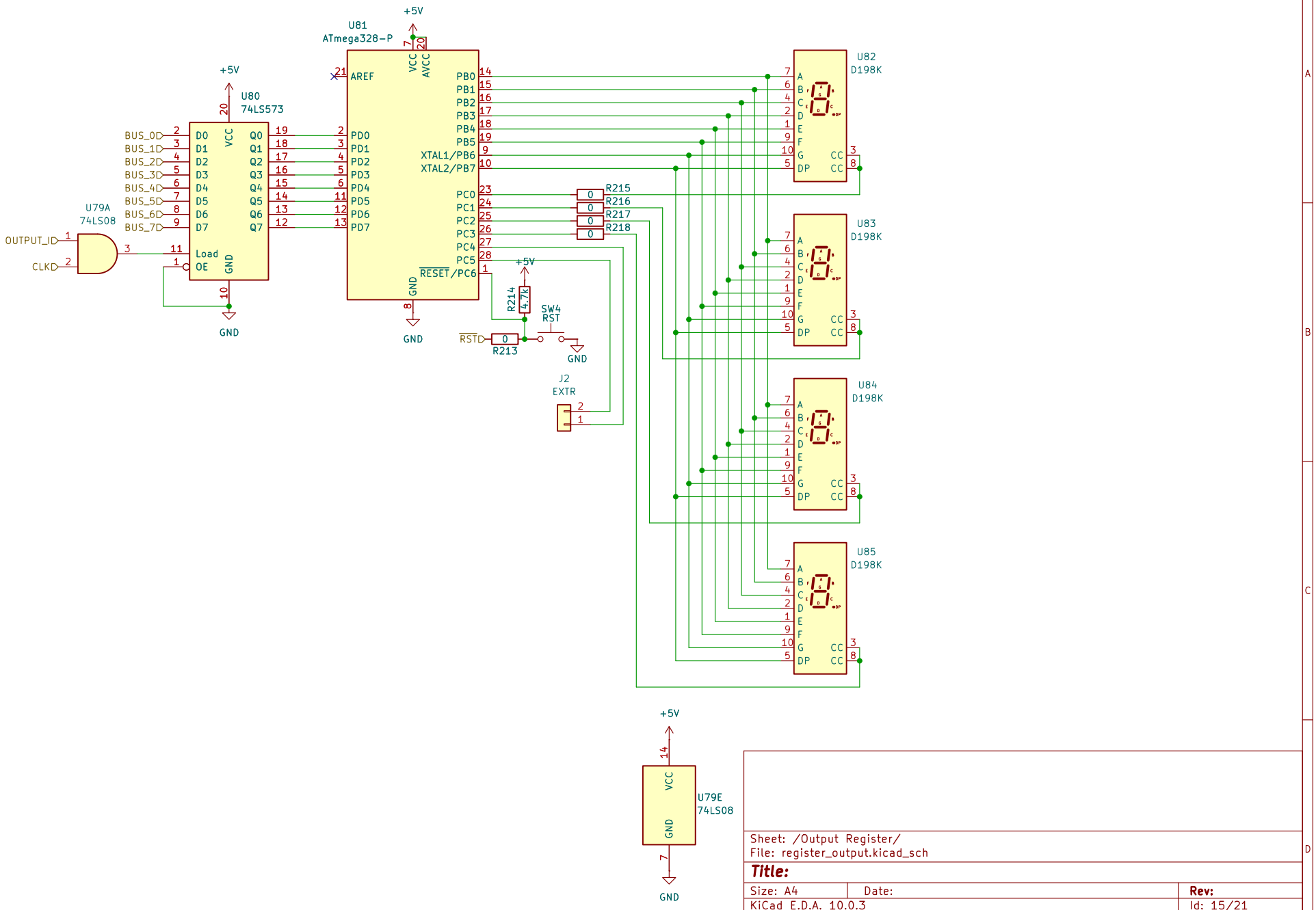
Sheet: /Register pair DE/
File: register_d_e.kicad_sch
Title:
Size: A3 Date: Rev:
KiCad E.D.A. 10.0.3 Id: 13/21

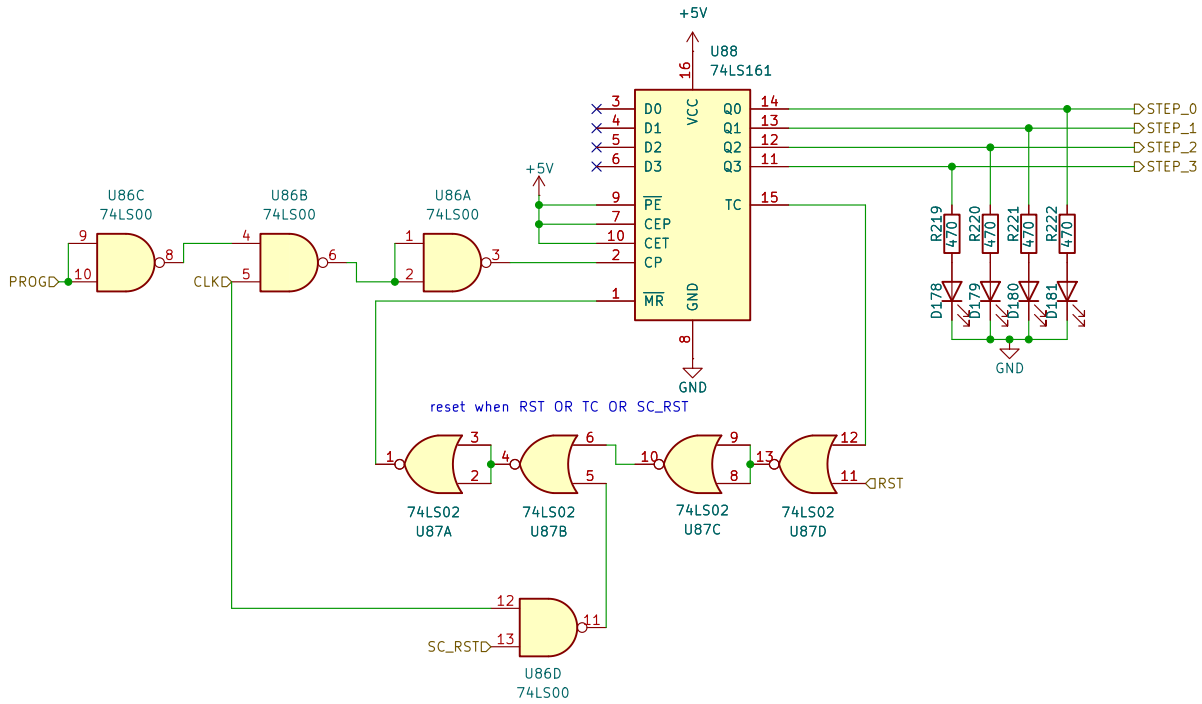


Sheet: /Instruction Register/
File: register_instruction.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.3 | Id: 14/21

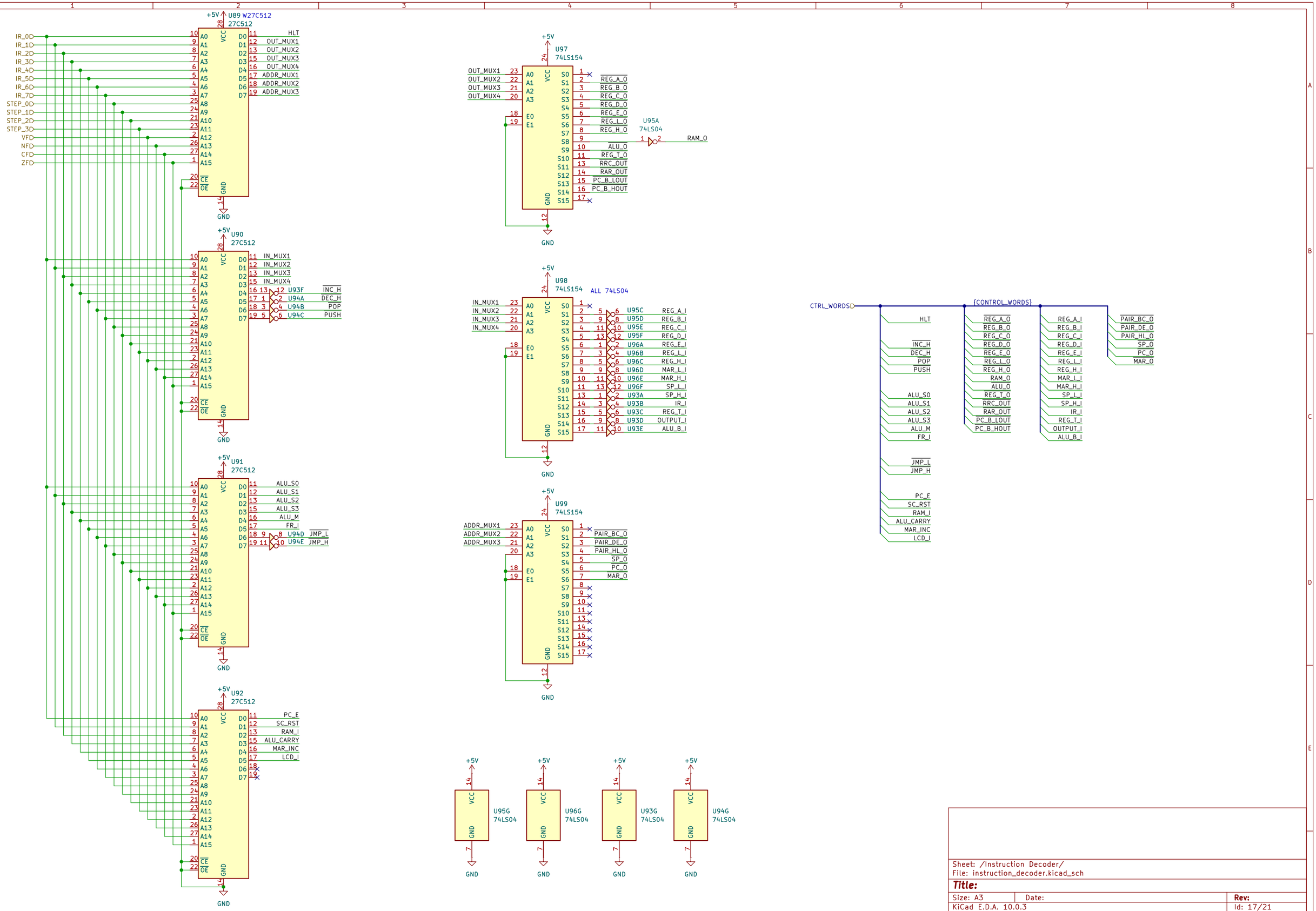

U81
ATmega328-P
+5V
AREF
VCC
AVCC
PB0
PB1
PB2
PB3
PB4
PB5
XTAL1/PB6
XTAL2/PB7
PC0
PC1
PC2
PC3
PC4
PC5
RESET/PC6
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
GND
U80
74LS573
BUS_0
BUS_1
BUS_2
BUS_3
BUS_4
BUS_5
BUS_6
BUS_7
D0
D1
D2
D3
D4
D5
D6
D7
Q0
Q1
Q2
Q3
Q4
Q5
Q6
Q7
Load
OE
U79A
74LS08
OUTPUT_ID
CLKD
R215
R216
R217
R218
R214
4.7k
SW4
RST
R213
RST
J2
EXTR
U82
U83
U84
U85
D198K
A
B
C
D
E
F
G
DP
CC
U79E
74LS08
Sheet: /Output Register/
File: register_output.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 15/21



Sheet: /Step Counter/
File: step_counter.kicad_sch

Title:

Size: A4 | Date: | **Rev:**
KiCad E.D.A. 10.0.3 | Id: 16/21

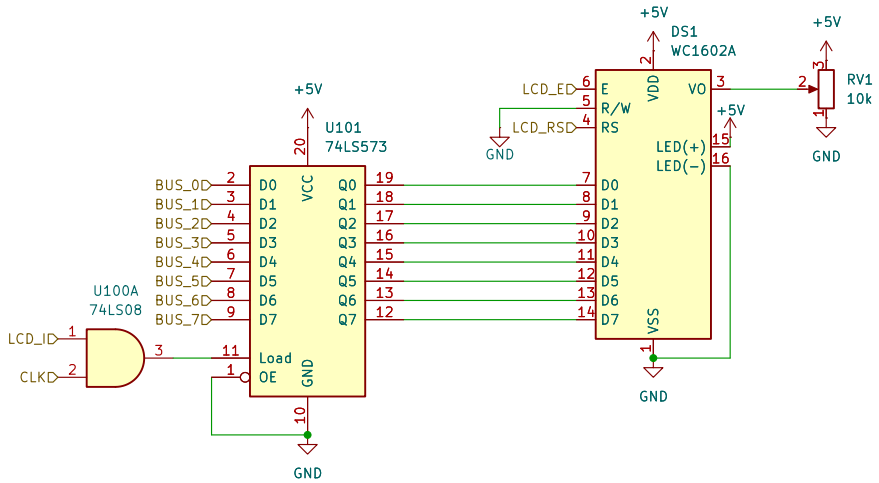


Sheet: /Instruction Decoder/
File: instruction_decoder.kicad_sch

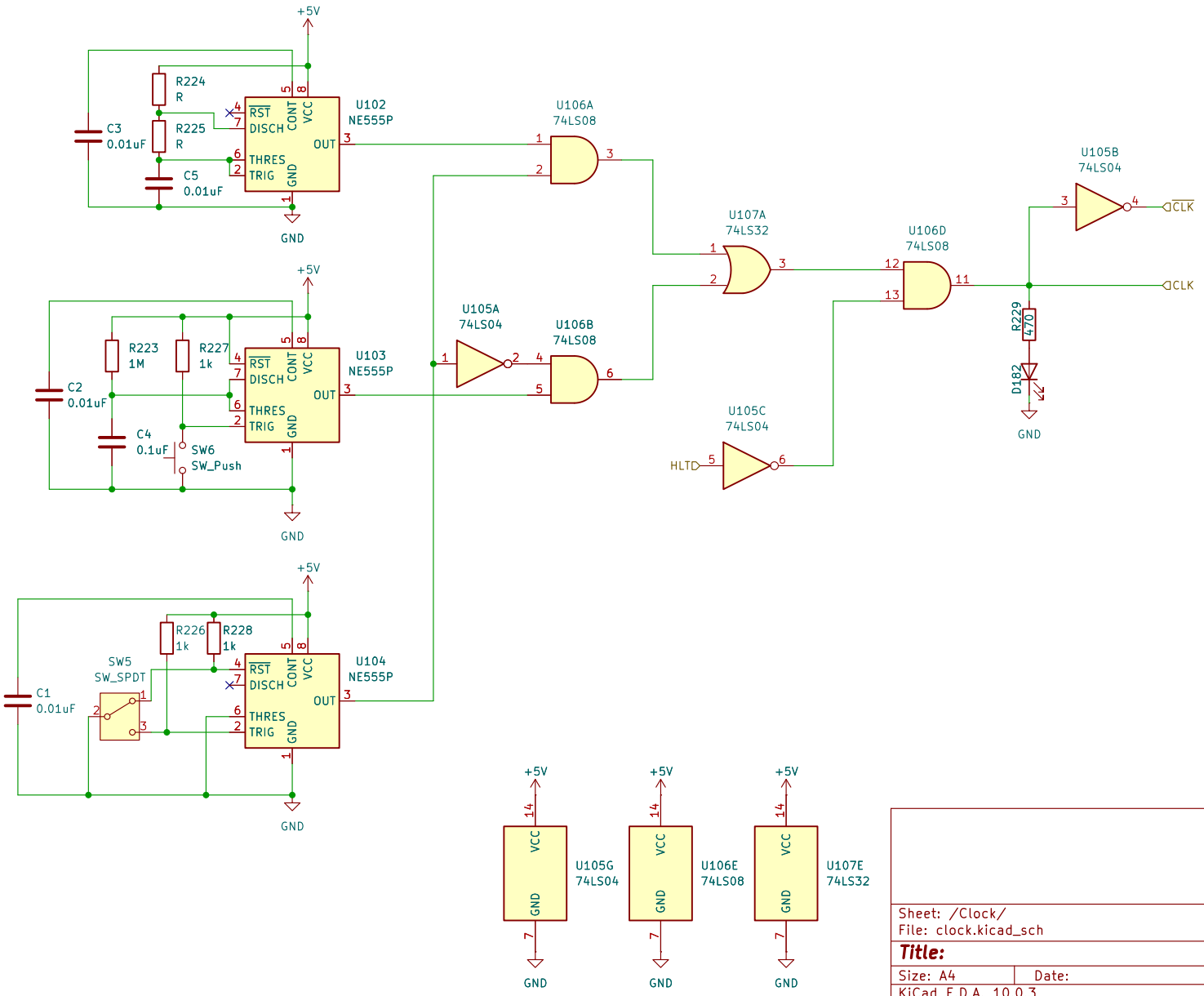
Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 10.0.3 | Id: 17/21



Sheet: /LCD Output/		
File: lcd_driver.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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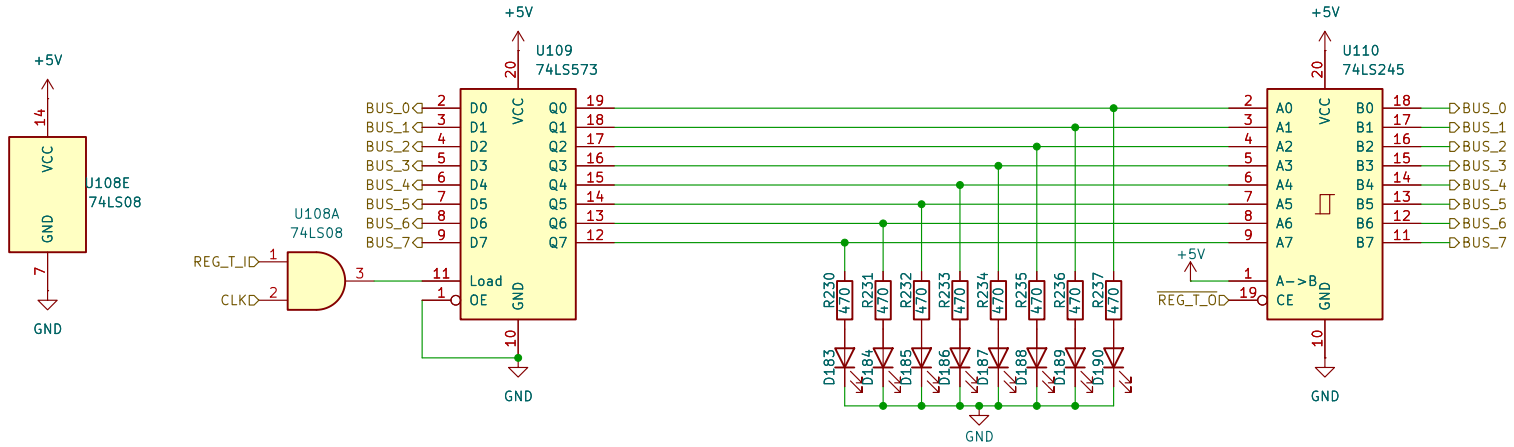


Sheet: /Clock/
File: clock.kicad_sch

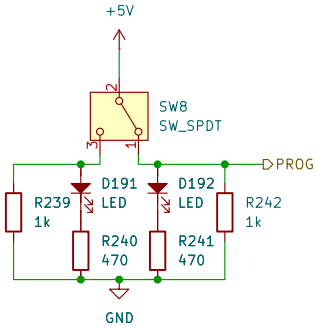
Title:

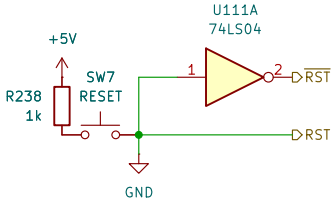
Size: A4 | Date: | **Rev:**

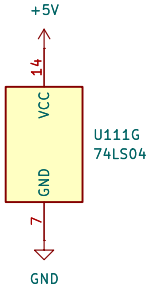
KiCad E.D.A. 10.0.3 | Id: 19/21



Sheet: /Temporary Register/		
File: register_temporary.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 20/21







Sheet: /Misc/
File: misc.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 21/21